

How can firms leverage responsible artificial intelligence to build competitive advantages? The role of supply chain justice and complexity

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ABSTRACT

In highly interconnected supply chains, firms are increasingly adopting artificial intelligence (AI) to improve their forecasting, resource allocation, and responsiveness. However, opaque and bias-prone algorithms can obscure decision-making processes and undermine collaboration between supply chain partners. Responsible AI, in which AI is designed, developed, and deployed in such a way that ensures its ethical conduct, transparency, and alignment with human values, offers a potential remedy, but its relational effects in supply chains remain unclear. Drawing on organizational justice theory, we test whether supply chain justice, encompassing both distributive and procedural justice, mediates the responsible AI–competitive advantage link, with supply chain complexity as a boundary condition. Survey data from 218 Chinese firms show that responsible AI fosters both distributive and procedural justice, which in turn facilitates firms' competitive advantages. Further, we identify an asymmetric moderating effect: high supply chain complexity weakens the mediating effect of distributive but not procedural justice. Our study advances the research on AI-enabled supply chains by identifying how ethical AI practices can foster firm performance via supply chain justice and when supply chain complexity weakens this effect. Practically, our findings suggest that in complex supply networks, ensuring transparent and consistent decision-making processes provides a more robust mechanism for sustaining competitive advantages compared with focusing only on outcome allocation.

1. Introduction

In today's rapidly evolving global markets, supply chains have become increasingly complex networks of interconnected firms (Lee et al., 2025) in which artificial intelligence (AI) plays a critical role in driving operational efficiency and strategic decision-making. AI enables firms to forecast their demand more accurately, optimize their resource allocation, and improve their responsiveness, creating significant competitive advantages (Benbya et al., 2021; Roberts and Candi, 2024). However, the widespread adoption of AI also introduces new relational challenges (Benbya et al., 2020; Fosso Wamba et al., 2023). The “black box” nature of AI systems and their potential embedded biases can obscure how decisions are made and raise concerns about fairness among supply chain partners (Agrawal et al., 2020). When partners

perceive decisions as unfair or opaque, this can erode trust, undermine collaboration, and diminish the expected firm performance benefits brought by investments in AI (Sartori and Bocca, 2023; Schoenherr et al., 2023).

In response to this challenge, the concept of “responsible AI” has emerged as a strategic approach to the design, development, and deployment of AI systems to ensure they operate ethically, promote social good, and align with fundamental human values (Dignum, 2019; Laine et al., 2024). Within supply chains, responsible AI involves partner-facing practices such as bias mitigation in AI-driven resource allocation and partner selection, explainable AI outputs that are accessible to stakeholders, and robust data protection mechanisms (Kumar et al., 2023). While responsible AI holds great promise, its outcomes will depend largely on how supply chain partners perceive and respond to

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the fairness of AI-driven decisions and processes. This brings into focus the crucial role of supply chain justice—the shared perception among partners that both the outcomes they receive (distributive justice) and the procedures used to reach those outcomes (procedural justice) are fair and equitable (Narasimhan et al., 2013). These perceptions are not peripheral; rather, they fundamentally shape how partners trust and engage with one another (Lee et al., 2025). However, while prior researchers have explored justice in traditional buyer–supplier contexts (Liu et al., 2012), whether responsible AI serves as an antecedent to interorganizational justice and, subsequently, competitive advantages remains a critical gap.

To address this gap, we draw on organizational justice theory (OJT), which posits that perceptions of fairness are fundamental determinants of organizational behavior and performance (Greenberg, 1987). Unlike purely economic views of exchange, OJT suggests that when partners perceive decision-making outcomes and processes as fair, they are less likely to engage in defensive behaviors and more likely to commit to long-term cooperation (Griffith et al., 2006). Drawing on this theoretical framework, we argue that responsible AI serves not only as a technical tool but also as a governance mechanism that signals fairness. With transparency, accountability, and equity embedded into its algorithms, responsible AI may improve supply chain partners' perceptions of distributive and procedural justice, in turn reducing transaction costs and fostering the necessary collaboration for competitive advantages.

However, these relationships do not exist in a vacuum. Supply chains operate in increasingly complex environments, characterized by multiple tiers of partners and fragmented information flows (Choi and Krause, 2006). Supply chain complexity (SCC) refers to the difficulties involved in managing and coordinating a diverse network of firms, processes, and relationships (Choi and Krause, 2006). From an OJT perspective, high complexity creates “noise” that may obscure the visibility of fairness signals (Colquitt, 2001). In complex networks, partners may struggle to verify whether an AI-driven outcome is truly equitable or a process is consistently applied (Wilhelm et al., 2016). Recognizing complexity as a critical boundary condition, we examine when and how it conditions the effectiveness of responsible AI. Accordingly, we propose two specific research questions:

RQ1: *Do distributive and procedural justice mediate the positive relationship between responsible AI and competitive advantages?*

RQ2: *Does SCC moderate the relationship between responsible AI and supply chain justice and, consequently, competitive advantages?*

We gather data from 218 Chinese firms actively engaged in AI implementation in their supply chain operations. The empirical results allow us to make three key contributions to the existing literature. First, we extend the research on AI value generation by moving beyond the binary view of AI adoption. While some scholars argue that the adoption of responsible AI can enhance user satisfaction (Wang et al., 2021) and organizations' market (Kumar et al., 2023) and environmental (Lin et al., 2024) performance, others caution against overestimating its effects, emphasizing the challenges involved in its implementation (Lee et al., 2023; Tehrani et al., 2024). We identify supply chain justice as a specific relational mechanism that translates responsible AI practices into competitive advantages. Second, we enrich the application of OJT in digital supply chains. By demonstrating that responsible AI acts as a distinct antecedent to both distributive and procedural justice, we provide a new theoretical explanation for how algorithmic governance influences partner behaviors and strategic performance. Third, we provide a nuanced understanding of SCC as a boundary condition. Contrary to the assumption that complexity universally degrades relationships, our findings reveal an asymmetry: high complexity weakens the signaling power of distributive justice (outcomes) but leaves procedural justice (process) unaffected. This offers a theoretical refinement regarding the robustness of process-based fairness in uncertain environments and provides managers with practical strategies for operating in complex

supply networks.

2. Literature review

2.1. Responsible artificial intelligence

The strategic importance of responsible AI can be examined through the lens of corporate social responsibility (Seelent et al., 2025). Studies show that ethical practices build intangible assets such as reputational capital and stakeholder trust, which translate into customer loyalty, better resource access, and superior financial performance, fostering sustainable competitive advantages (Branco and Rodrigues, 2006). Responsible AI represents an extension of these ethical commitments into the realm of digital governance. By embedding principles of accountability, responsibility, and transparency into the design and deployment of AI tools (Dignum, 2020), firms can signal their commitment to ethical conduct in their technological operations (Li et al., 2025). Recent research highlights that transparency, in particular, is critical for establishing legitimacy and reducing uncertainty in dyadic supply chain relationships (Liu et al., 2024). Consequently, the responsible governing of AI through these principles serves to mitigate substantial business risks, including reputational harm stemming from biased outcomes (Obermeyer et al., 2019), legal exposure from privacy violations (Cheng et al., 2022), and diminished partner trust arising from opaque “black box” decision-making (Haibe-Kains et al., 2020).

Despite these potential benefits, empirical evidence on the business value of adopting responsible AI practices remains inconclusive (Hendriksen, 2023). While some studies show that responsible AI enhances user engagement in health care (Wang et al., 2021), strengthens risk management capabilities (Renieris et al., 2022), and positively influences market performance through customer value cocreation (Kumar et al., 2023), others caution that these benefits are not guaranteed (Aker et al., 2022; Lee et al., 2023). The considerable financial investment, specialized expertise, and organizational transformation needed for the implementation of responsible AI can present formidable barriers, particularly for resource-constrained small and medium-sized enterprises (De Bruyn et al., 2020; Wamba and Queiroz, 2023).

This tension underscores a critical gap in the literature. To reconcile these mixed findings, research must move beyond the binary question of whether adopting responsible AI leads to competitive advantages and instead investigate the mechanisms that translate internal commitments into externally recognized benefits. In particular, in the context of complex and interdependent supply chain relationships, the pathways by which responsible AI creates relational value remain underexplored (Makarius et al., 2020). Merely deploying a technically “responsible” AI tool does not guarantee positive outcomes if supply chain partners do not perceive its operations and results as fair and transparent. This suggests that the value of responsible AI may hinge on relational mechanisms that shape partners' perceptions and behaviors. Therefore, understanding how responsible AI influences perceptions of interorganizational justice within the supply chain is crucial to explaining its contribution to sustainable competitive advantages.

2.2. Supply chain justice

Justice is commonly defined as the fair and equitable treatment of individuals and groups (Colquitt, 2001). Because supply chains often involve multiple stakeholders, including suppliers, manufacturers, and distributors (Bozarth et al., 2009), scholars are increasingly recognizing the importance of justice and exploring how different types of supply chain justice apply specifically to partnerships (Porter and Kramer, 2006). Distributive justice refers to the perceived fairness of resource allocation and outcomes (Adams, 1965). Procedural justice focuses on the fairness of the processes and decision-making procedures that lead to these outcomes (Leventhal, 1980).

A growing body of work has investigated the effects of various types

of supply chain justice on interorganizational relationships and performance (Griffith et al., 2006; Liu et al., 2012; Qiao et al., 2022). For example, in a study of 3225 wholesale distributors, Griffith et al. (2006) found that both procedural and distributive justice were positively associated with buyer-supplier relationship quality, ultimately improving performance. Similarly, based on data from 402 Chinese dyads, Y. Liu et al. (2012) demonstrated that supply chain justice enhances buyer-supplier relationship performance by strengthening knowledge sharing, commitment, and relationship investment. More recently, Qiao et al. (2022) highlighted the moderating role of supplier-perceived justice, which positively influences the effect of relationship attractiveness on environmental collaboration and supplier environmental commitment. Together, these studies illustrate the significant role of supply chain justice in the dynamics of interorganizational relationships and driving both economic and non-economic performance outcomes.

While prior studies provide valuable insights (Liu et al., 2012), a critical gap remains in linking supply chain justice to the rapidly evolving field of responsible AI. Earlier justice work predates the widespread adoption of AI, which is now guided by accountability, responsibility, and transparency principles to enhance stakeholder trust and acceptance (Kumar et al., 2023). More importantly, these principles should directly influence distributive and procedural justice. Distributive justice relates to the fairness of the AI-mediated allocation of benefits and burdens, such as pricing, inventory prioritization, or risk sharing (Ziari and Taleizadeh, 2025). Procedural justice relates to the fairness of AI-driven decision-making processes, including supplier selection, contract evaluation, and demand forecasting (Prashar, 2025). These two dimensions are emphasized because they capture the most direct and observable ways in which responsible AI principles manifest in interorganizational exchanges, shaping both outcomes and processes. In other words, understanding the effect of responsible AI on these two types of justice is key to uncovering its business value.

2.3. Supply chain complexity

Supply chain complexity (SCC) refers to the difficulties of managing and coordinating a diverse network of firms, processes, and relationships within a supply chain (Choi and Krause, 2006). Because SCC reflects a firm's operating environment (Bode and Wagner, 2015), researchers have examined its moderating role. For example, using data from 145 companies from Spain and the Netherlands, Gimenez et al. (2012) found that the performance benefits of supply chain integration are amplified in high-complexity supply chains but less pronounced in simpler contexts. In contrast, analyzing data from 1161 German firms, Blome et al. (2014) found that SCC positively moderates the relationship between external knowledge transfer and flexibility and negatively moderates the relationship between internal knowledge transfer and flexibility.

Despite the extensive research on SCC, its effect on social and ethical dynamics, particularly on the implementation of responsible AI, has received limited scholarly attention. In complex, multitiered supply chains, information flow often becomes fragmented and opaque (Mackelprang and Malhotra, 2015). As a result, a focal firm's responsible AI initiatives may lose clarity and influence when transmitted to Tier 2 and Tier 3 suppliers (Wilhelm et al., 2016). This diminished visibility can foster suspicion and reduce stakeholders' ability to accurately perceive fairness. Further, the logistical challenges of uniformly applying responsible AI principles increase significantly with the size and diversity of the supply network (Busse et al., 2017). Overall, neglecting this critical interaction carries the risk of overestimating the benefits of responsible AI because structural fragmentation and operational complexities may undermine efforts to ensure fairness across the supply chain. Therefore, SCC should be considered not merely an independent factor but a pivotal moderator that determines whether responsible AI can successfully foster justice perceptions necessary for

sustainable competitive advantages.

3. Theory and hypotheses

3.1. Organizational justice theory

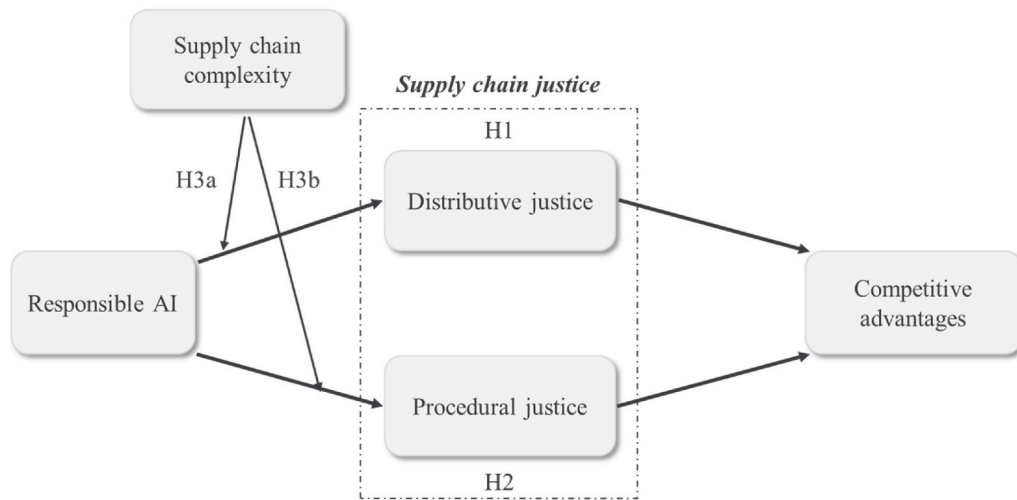
Organizational justice theory (OJT) posits that the perception of fairness with respect to resource allocation and decision-making processes is a fundamental determinant of individual and group behavior in organizational settings (Greenberg, 1987). While initially developed to explain employee attitudes, OJT has been widely extended to supply chain management to understand interorganizational dynamics (Griffith et al., 2006; Liu et al., 2012). In the context of supply chain relationships, interorganizational justice refers to the extent to which a firm perceives that it is being treated fairly by its partners in terms of both outcomes and procedures (Alghababsheh et al., 2023). OJT suggests that when partners perceive justice, they are more likely to engage in cooperative behaviors, reduce opportunistic actions, and commit to long-term relationship stability (Kaynak et al., 2015), all of which are critical drivers of competitive advantages (Branco and Rodrigues, 2006).

Guided by OJT, we theorize a moderated mediation pathway from responsible AI to competitive advantages via supply chain justice (see Fig. 1). Specifically, by embedding principles of equity, transparency, and ethicality into AI systems, firms may strengthen their partners' perceptions of distributive justice (fairness of outcomes) and procedural justice (fairness of processes), in turn fostering relations conducive to efficiency and innovation, thereby enhancing competitive advantages. Our model focuses specifically on distributive and procedural justice because they capture the most direct structural effects of responsible AI on supply chain exchanges. Responsible AI fundamentally alters what partners receive through algorithmic allocation (distributive justice) and how those decisions are computed through standardized protocols (procedural justice). Finally, we position SCC as a boundary condition that alters the visibility and interpretability of these justice signals.

3.2. Mediating role of supply chain justice

Distributive justice refers to the extent to which a supply chain partner perceives that the returns from its inputs into the relationship are essentially equal to those of the other party (Alghababsheh et al., 2023). Central to the conceptualization of distributive justice is the principle of equity (Adams, 1965), in which firms assess fairness by comparing their own outcomes-to-inputs ratio with that of their partners or peers. When this ratio is perceived as equal, equity is established. Further, distributive justice encompasses needs-based allocation, where outcomes are distributed according to a firm's needs, regardless of its contributions, preserving social relations (Deutsch, 1975). We posit that responsible AI significantly fosters perceptions of distributive justice by optimizing equity and needs-based allocations.

First, in demand forecasting and inventory management, responsibly designed AI can optimize resource allocation based on objective data (i. e., historical sales and capacity) rather than subjective favoritism, increasing the perceived consistency and fairness of allocations. By precisely balancing supply with demand, responsible AI ensures that partners receive an appropriate share of resources commensurate with their contributions (Seyedan and Mafakheri, 2020). The objective alignment of inputs (partner contributions) and outcomes (resource allocation) satisfies the core equity criterion of distributive justice. Second, in pricing mechanisms, responsible AI implements transparent algorithms that link prices directly to clear cost structures and market conditions. This visibility enables partners to trace the logic behind economic decisions and perform the necessary social comparisons to verify that margins are distributed fairly (Huo et al., 2015). Such transparency reduces suspicions of negative equity (i. e., being exploited) and reinforces the belief that financial outcomes are grounded in verifiable market realities. Third, with respect to payments, responsible



Note: H1 and H2 predict mediation of the relationship between responsible AI and competitive advantages, with distributive justice and procedural justice as the respective mediators. H3a and H3b predict moderation of the same mediation relationships by supply chain complexity.

Fig. 1. Research framework.

AI facilitates accurate and timely disbursements, which addresses needs-based justice (Deutsch, 1975). By reducing cash flow uncertainty, the focal firm demonstrates a commitment to the partner's financial stability (Kordzadeh and Ghasemaghaei, 2022), signaling that the distribution of financial risks is shared equitably rather than offloaded onto the supplier.

According to OJT, these enhanced perceptions of distributive justice translate into competitive advantages by reducing relational frictions and motivating cooperative resource sharing. When partners perceive equity, they are less inclined to engage in costly bargaining or defensive monitoring to protect their interests, thereby lowering transaction costs. More importantly, perceived fairness motivates partners to voluntarily benefit the focal firm. For example, during the post-pandemic recovery, automobile manufacturers that consistently applied fair allocation principles to their suppliers received priority access to critical components, helping them to directly strengthen their operational resilience and market share (Chen et al., 2022). Similarly, JD.com's use of intelligent platforms to ensure fair pricing and equitable fulfillment during COVID-19 not only secured partner cooperation but also enhanced consumer loyalty and repurchase rates, directly influencing revenue growth and brand competitiveness (Shen and Sun, 2023).

In sum, responsible AI indirectly fosters a firm's competitive advantages by cultivating its partners' perceptions of distributive justice. Therefore, we propose:

H1. Distributive justice mediates the positive association between responsible AI and competitive advantages.

Procedural justice refers to the degree to which a supply chain partner perceives that the procedures, policies, and decision-making processes governing the relationship are fair (Alghababsheh et al., 2020). According to Leventhal's (1980) framework, procedural fairness is based on specific structural criteria, most notably consistency, unbiasedness, and representativeness (voice). We posit that responsible AI fosters procedural justice not only by automating tasks but also by embedding these specific fairness criteria into supply chain governance.

First, responsible AI strengthens consistency and unbiasedness by replacing opaque or arbitrary manual decisions with transparent, rules-based algorithms. Traditional decision-making is prone to human error and ad hoc changes, creating uncertainty. In contrast, responsible AI helps ensure that decision rules (e.g., regarding supplier selection or logistics routing) are applied consistently across different times and

partners (Schoenherr et al., 2023). Further, by incorporating explainable AI, responsible AI articulates the rationale behind these decisions (Arrieta et al., 2020), allowing partners to verify that the "rules of the game" are stable, predictable, and free from hidden biases. This combination of consistency and transparency satisfies the core requirement of procedural regularity. Second, responsible AI fosters representativeness (or "voice") by embedding stakeholder participation into the system. Mechanisms that enable partners to query AI outputs or appeal decisions signal that their perspectives are valued. For instance, Deutsche Telekom's inclusion of stakeholder representatives in AI governance ensures that decision parameters reflect broader interests (Haipeter et al., 2024). From an OJT perspective, this participatory process transforms decision-making from a unilateral imposition into a collaborative governance structure.

In turn, high procedural justice drives competitive advantages by influencing partners' attitudes and behaviors in ways that reduce costs and enhance value. First, procedural justice significantly reduces opportunistic behaviors and conflict (Brown et al., 2006; Luo et al., 2015). When partners believe that decision-making rules are consistent and unbiased (e.g., automated by AI), their perceived need to engage in defensive monitoring or costly safeguards against potential exploitation decreases. The reduction in friction and conflict resolution lowers overall transaction costs, directly contributing to the focal firm's cost advantage (Narasimhan et al., 2013). Second, procedural justice fosters relationship-specific investment and knowledge sharing (Huo et al., 2016; Liu et al., 2012). According to OJT, fair procedures signal long-term orientation and commitment (Hofer et al., 2012), encouraging partners to invest in specialized assets and share proprietary information without fear of misappropriation. For example, Ling-yee (2010) found that procedural justice drives voluntary cooperation and knowledge exchange. These collaborative behaviors enable the focal firm to innovate faster, improve product quality (Carnovale et al., 2019), and respond more agilely to market changes, thereby strengthening its competitive advantages.

In sum, empirical evidence supports a direct link between procedural justice and strategic performance (J. Hoppner et al., 2014). By institutionalizing fair processes through responsible AI, firms can cultivate perceptions of procedural justice needed to not only secure compliance but also actively unlock the operational synergies that underpin sustained competitive advantages. Therefore, we propose:

H2. Procedural justice mediates the positive association between

responsible AI and competitive advantages.

3.3. Moderating role of supply chain complexity

The process through which responsible AI fosters supply chain justice and, subsequently, competitive advantages depend on the context. Because SCC shapes the structural and operational environment in which partners interpret and respond to responsible AI initiatives, we posit that it acts as a critical boundary condition (Bozarth et al., 2009; Choi and Krause, 2006). From an OJT perspective, the formation of justice perceptions relies heavily on the visibility and clarity of the treatment received (Colquitt, 2001). However, high complexity introduces environmental noise and information asymmetry (Wilhelm et al., 2016), which may distort or dilute the fairness signals generated by responsible AI, thereby altering the effectiveness of the justice mechanism.

In the case of distributive justice, OJT posits that partners assess the fairness of outcomes through a process of social comparison, meaning they compare their input-to-outcome ratio with that of others (Adams, 1965). High SCC undermines this comparison process by introducing severe informational opacity (Wang et al., 2022). In a multitier, highly interconnected network, the link between the focal firm's AI-driven decisions (e.g., resource allocation) and the final outcome experienced by a partner is often obscured by intermediaries or network fluctuations. This attributional ambiguity makes it difficult for partners to verify whether an allocation is truly based on equitable principles or is a result of systemic randomness. When partners cannot confidently attribute outcomes to the focal firm's fairness, the positive effect of responsible AI on distributive justice is weakened. Consequently, the ability of distributive justice to mediate competitive advantages is diminished because the equity signal is lost in the noise of complexity. Therefore, we propose:

H3a. High SCC negatively moderates the mediating role of distributive justice in the relationship between responsible AI and competitive advantages.

Similarly, for procedural justice, high SCC challenges Leventhal (1980) criteria, particularly consistency and accuracy. While responsible AI aims to deliver consistent and transparent decision-making, high SCC imposes significant barriers to the consistent delivery of these procedures across a diverse and dispersed network (Ho et al., 2015). In complex supply chains, partners possess varying levels of technological capability and cultural understanding. Presenting AI-driven explanations in a form that is uniformly accessible and accurate for all stakeholders is logistically challenging (Busse et al., 2017). Further, mechanisms for stakeholder voice, a key element of procedural justice, may become overwhelmed. As the number of network nodes increases, soliciting, integrating, and responding to feedback from all partners becomes increasingly difficult (Dukes and Streeck, 2022). This potential for communication overload and inconsistent implementation can erode partners' perceptions that the process is truly fair and inclusive. Consequently, we anticipate that in highly complex environments, the mediating effect of procedural justice on the relationship between responsible AI and competitive advantages could be weakened. Therefore, we propose:

H3b. High SCC negatively moderates the mediating role of procedural justice in the relationship between responsible AI and competitive advantages.

4. Methods

4.1. Data collection

For our empirical setting, we selected China's manufacturing and information technology (IT) sectors, which are among the most AI-

intensive sectors in China's supply chain operations, with recent evidence showing widespread deployment of AI in supplier selection, order allocation, dynamic pricing, logistics planning, and risk monitoring (World Economic Forum, 2025). Importantly, these sectors differ systematically in their supply chain network positions and technological roles: manufacturing firms serve as the core of physical flows and production coordination, while IT firms act as both adopters and providers of the AI infrastructure that underpins digital supply chains (Maslej et al., 2025). Studying both sectors provides a broader test of the universality and boundary conditions of our theorized relationships. Finally, China offers a timely and policy-relevant context. Since 2023, regulators have issued detailed rules on algorithmic transparency, bias mitigation, explainability, and privacy (Cyberspace Administration of China, 2023), making the responsible use of AI critically important in supply chain operations.

We employed a rigorous, multistage process to develop and validate our survey instrument. First, the survey items were translated from English to Chinese by a bilingual academic expert in operations management, then translated back to English by another bilingual academic expert in supply chain management. Discrepancies were discussed and resolved, with minor wording adjustments made for cultural appropriateness while preserving the original meaning. To ensure contextual relevance, we conducted an expert review with two senior industry professionals with over 10 years' experience in AI-enabled supply chain management. These experts were identified and invited through one coauthor's professional network, ensuring both expertise and willingness to provide in-depth feedback. Their input led to minor adaptations to reflect the differences between Western and Chinese business environments.

We then conducted a pretest with a professional survey provider using a small sample of managers with experience in supply chain or technology roles from firms similar to our target population. Feedback from the pretest led to further refinements to improve clarity and reduce ambiguity. Finally, our target population comprised senior managers (e.g., vice presidents, directors, and senior managers) in Chinese manufacturing and IT firms who have actively adopted AI in their supply chain operations. Firms were excluded if they were only considering AI adoption, had made no AI investments in the past 3 years, or had no plans to expand AI usage in the next 3 years. We invited 335 qualified firms from the survey provider's database and received 218 valid responses after applying exclusion criteria and quality checks (abnormal completion time, missing data, or patterned answers), resulting in an effective response rate of 65.07%. Table 1 shows the sample characteristics.

4.2. Measures

All constructs in our study were adapted from well-established measures in the prior literature. First, responsible AI was measured

Table 1
Demographic profile (N = 218).

Characteristic	No.	%
Number of employees (firm size)	≥1000	42
	300–999	89
	≤299	87
Years of existence (firm age)	<6 years	5
	6–10 years	44
	11–15 years	57
	16–20 years	75
	>20 years	37
Industry type	Manufacturing	112
	Information technology	87
	Other	19
Ownership	State owned	23
	Privately owned	172
	Other	23

using four items adapted from P. Kumar et al. (2023). We selected this scale because it operationalizes responsible AI via three aspects—technical skills, ethical concerns, and risk mitigation—which together embody the principles of accountability, responsibility, and transparency (Dignum, 2020).

Second, for supply chain justice, we distinguished between distributive and procedural justice. Distributive justice was measured using three items from Y. Liu et al. (2012), who built on the foundational work of N. Kumar et al. (1995) and Luo (2007). This scale was chosen because it is not only theoretically established but has also been empirically validated in the Chinese manufacturing sector (Liu et al., 2012). The scale captures partners' holistic assessment of relationship quality rather than their reaction to isolated technical events. This ensures the construct reflects the strategic relational effect of AI governance. Procedural justice was measured using three items adapted from Narasimhan et al. (2013), selected for their strong conceptual grounding in capturing the key elements of fair process: fairness in interactions, clarity in communicating decision criteria, and consistency in applying those criteria.

Third, competitive advantages were assessed using three items from L. Li et al. (2023) that benchmark financial performance against industry peers. This measure was chosen for its ability to capture relative market positioning, which is consistent with our focus on performance implications in competitive supply chain environments.

Finally, for SCC, we adapted three items from Bode and Macdonald (2017). While some measures define complexity purely in structural terms (e.g., the number of nodes or members), our study required a broader, perceptual measure because responsible AI adoption responds to overall perceived complexity, not just network size. This scale captures managerial perceptions of how difficult the supply chain is to explain, understand, or analyze, making it more appropriate for examining strategic responses across diverse firms.

4.3. Bias statements

To mitigate potential nonresponse bias, we implemented several strategies. First, the questionnaire underwent iterative refinement through cognitive pretesting and expert review to eliminate ambiguous phrasing, reduce item complexity, and mitigate the effects of question order. Additionally, response patterns were systematically monitored via temporal analysis, with particular attention paid to response latency metrics. Finally, a comparative analysis between early respondents (first quartile) and late respondents (final quartile) was conducted using Levene's test for homogeneity of variance (Armstrong and Overton, 1977), revealing no statistically significant differences in response distributions between the two groups ($p > .05$).

To address common method bias, we adopted a multistage procedural approach. The survey design incorporated randomized item sequencing and scale polarity reversal during instrument development to disrupt response pattern artifacts. Moreover, participants received standardized instructions with assurances of confidentiality and disassociation to reduce evaluation apprehension. Further, Harman's single-factor test demonstrated poor model fit when constraining all items to a single latent factor ($\chi^2 = 850.249$, $df = 104$, $\chi^2/df = 8.175$, goodness-of-fit index [GFI] = 0.667, Tucker–Lewis index [TLI] = 0.449,

comparative fit index [CFI] = 0.523, incremental fit index [IFI] = 0.527, root mean square error of approximation [RMSEA] = 0.182) (Li et al., 2019). Finally, a theoretically unrelated marker variable ("Please rate your family's happiness", scored on a seven-point Likert scale) was incorporated following the partial correlation procedure (Lindell and Whitney, 2001). As shown in Table 2, the marker variable exhibited negligible correlations with substantive constructs, providing empirical evidence that common method bias did not constitute a substantive threat.

5. Results

5.1. Reliability and validity

To assess the reliability and validity of the measurement instruments, we conducted confirmatory factor analysis of all five latent constructs: responsible AI, distributive justice, procedural justice, competitive advantages, and SCC. The results indicate that the model fit the data well ($\chi^2 = 172.999$, $df = 94$, $\chi^2/df = 1.840$, GFI = 0.909, TLI = 0.935, CFI = 0.949, IFI = 0.950, RMSEA = 0.062). The analysis also yielded standardized factor loadings for each item on its respective construct. As shown in Fig. A1, all factor loadings were statistically significant and ranged from 0.633 to 0.909, indicating that each item made a strong contribution to the measurement of its underlying construct (Fornell and Larcker, 1981). We then assessed internal consistency and convergent validity. Internal consistency was evaluated using composite reliability and Cronbach's alpha. As detailed in Fig. A1, the composite reliability values for all constructs ranged from 0.799 to 0.879, surpassing the recommended threshold of 0.70 (Bentler, 1990). Similarly, Cronbach's alpha values ranged from 0.796 to 0.876, also exceeding the 0.70 threshold (Nunnally and Bernstein, 1978). Convergent validity was evaluated using average variance extracted (AVE). The AVE values for all constructs ranged from 0.521 to 0.709, surpassing the recommended threshold of 0.50 (Bentler, 1990). This suggests that the items within each construct share a substantial proportion of common variance (Podsakoff et al., 2003). Finally, as shown in Table 2, the square root of the AVE for each construct was greater than pairwise correlations with other constructs, demonstrating that each construct was distinct from the others, providing evidence of discriminant validity (Podsakoff et al., 2003).

5.2. Hypothesis testing

Before testing our hypotheses, we checked for the presence of multicollinearity among the predictor variables in our models. Variance inflation factor analysis measures the extent to which the variance of a regression coefficient is increased as a result of multicollinearity with other variables (Joseph et al., 2022). The results of our analysis revealed that all values were well below the commonly recommended threshold of 10 (Neter et al., 1996). Hence, multicollinearity was not a significant concern in our analysis, ensuring the reliability of our regression models and the validity of our hypothesis testing.

To test our hypotheses, we used the PROCESS macro in SPSS 23.0 (Hayes, 2014), with responsible AI as the independent variable, distributive justice and procedural justice as the mediators, SCC as the

Table 2
Mean, standard deviation, and correlations.

Variable	Mean	SD	1	2	3	4	5
1 Responsible AI	5.926	0.653	0.722				
2 Distributive justice	5.885	0.652	0.438**	0.755			
3 Procedural justice	5.766	0.776	0.475**	0.554**	0.778		
4 Competitive advantages	5.468	0.826	0.322**	0.482**	0.429**	0.776	
5 Supply chain complexity	4.621	1.219	0.076	0.100	0.036	0.060	0.842
6 Marker variable	NA	NA	−0.082	−0.024	−0.018	0.092	0.080

Note: AI: artificial intelligence. The diagonal displays the square roots of the average variance extracted. ** $p < .01$.

moderator, and competitive advantages as the dependent variable. Firm age, firm size, industry sector, and respondent position were included as control variables.

Table 3 presents the results of the regression models. Models 1 and 3 indicate that responsible AI is positively and significantly associated with distributive justice ($\beta = 0.412, p < .001$) and procedural justice ($\beta = 0.560, p < .001$), respectively. These findings suggest that organizations with a greater commitment to responsible AI practices are likely to experience higher levels of justice in both the distribution of resources (distributive justice) and the processes that govern decision-making (procedural justice).

Model 5, which estimates the direct (unmediated) relationship between responsible AI and competitive advantages, shows that responsible AI is positively associated with competitive advantages ($\beta = 0.395, p < .001$). However, when distributive and procedural justice are included as mediators in Model 6, the direct positive relationship between responsible AI and competitive advantages becomes nonsignificant ($\beta = 0.091, p > .05$). This suggests that the initial direct effect of responsible AI on competitive advantages is mediated by perceptions of justice within the organization. Specifically, distributive justice ($\beta = 0.399, p < .001$) and procedural justice ($\beta = 0.249, p < .001$) both show a significant positive relationship with competitive advantages, supporting the hypotheses that responsible AI indirectly enhances competitive advantages through these two forms of justice. Further, the results from Models 2 and 4 reveal the moderating effect of SCC. The interaction terms in Model 2 show that SCC significantly and negatively moderates the relationship between responsible AI and distributive justice ($\beta = -0.233, p < .001$).

To further interpret the nature of this moderating effect, we plotted simple slopes at high (+1 standard deviation) and low (−1 standard deviation) levels of SCC (see Fig. 2). The results of the simple slope analysis reveal that the positive relationship between responsible AI and distributive justice is stronger and highly significant when SCC is low (simple slope = 0.591). However, when SCC is high, this relationship, while still positive, becomes weaker (simple slope = 0.024). This pattern supports H3a, indicating that as SCC increases, the positive effect of responsible AI on distributive justice is weakened. In contrast, SCC did not moderate the relationship between responsible AI and procedural justice, indicating that it has no influence on the effect of responsible AI on procedural justice in the organization. Finally, to ensure that our main conclusions are not affected by potential data distribution issues (e.g., heteroscedasticity), we reestimated the models using heteroscedasticity-consistent standard errors (HC3), and the results

Table 3
Regression estimates.

Variable	Distributive justice		Procedural justice		Competitive advantages	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	3.368***	−2.727*	2.611***	−0.151	2.654***	0.659
<i>Control variables</i>						
Firm size	0.039	0.036	−0.104	−0.106	0.037	0.048
Firm age	−0.028	−0.017	−0.023	−0.018	0.120*	0.136**
State owned	0.041	0.032	0.221	0.212	0.275	0.204
Privately owned	−0.136	−0.136	0.102	0.099	−0.108	−0.080
Manufacturing firm	0.273	0.286	0.018	0.025	0.005	−0.109
Information technology firm	0.158	0.227	−0.031	0.001	0.128	0.072
<i>Independent variable</i>						
Responsible AI	0.412***	1.382***	0.560***	1.006***	0.395***	0.091
<i>Mediating variables</i>						
Distributive justice						0.399***
Procedural justice						0.249***
<i>Moderating variables</i>						
SCC		1.442***		0.654		
Responsible AI × SCC		−0.233***		−0.107		
df	7, 210	9, 208	7, 210	9, 208	7, 210	9, 208
R ²	0.217	0.298	0.243	0.255	0.159	0.330
F	8.323***	9.801***	9.623***	7.888***	5.657***	11.367***

Note. AI: artificial intelligence; SCC: supply chain complexity. *** $p < .001$; ** $p < .01$; * $p < .05$.

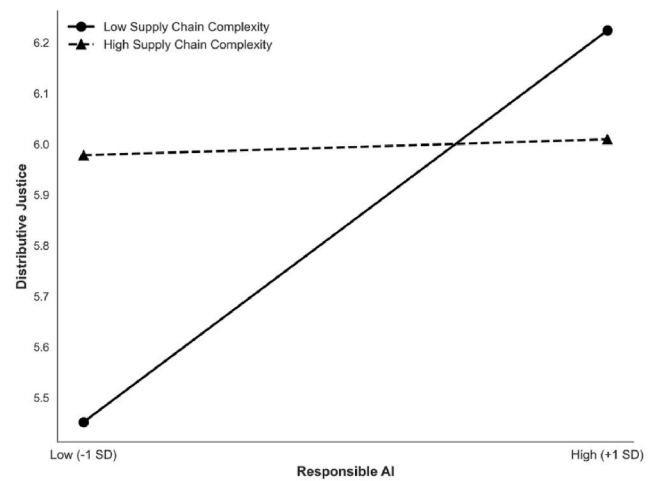


Fig. 2. Moderating effect of supply chain complexity on the relationship between responsible AI and distributive justice.

remained consistent with the main analysis.

Table 4 presents the findings of the mediation analysis using a bootstrapping approach. The statistical significance of the path coefficients was determined by checking that the confidence interval (CI) did not include zero. Specifically, both the total and indirect effects of responsible AI on competitive advantages were statistically significant, supporting the hypothesis that responsible AI indirectly influences competitive advantages via distributive and procedural justice. Further, the results of the moderated mediation analyses reveal different patterns for distributive and procedural justice. For distributive justice, the moderated mediation index is -0.093 ($SE = 0.036$, 95% CI $[-0.166, -0.028]$). This indicates that SCC significantly moderates the mediating role of distributive justice in the responsible AI–competitive advantage

Table 4
Indirect effects of responsible AI on competitive advantages.

Mediating variable	Effect	SE	95% CI
Aggregate	0.304	0.073	[0.167, 0.454]
Distributive justice	0.164	0.059	[0.060, 0.289]
Procedural justice	0.140	0.053	[0.035, 0.243]

relationship. In contrast, for procedural justice, the moderated mediation index is -0.027 ($SE = 0.020$, 95% CI $[-0.068, 0.013]$), which is not statistically significant. Based on these findings, we conclude that **H1** (the mediation effect of distributive justice), **H2** (the mediation effect of procedural justice), and **H3a** (the moderated mediation effect of distributive justice) are supported, while **H3b** (the moderated mediation effect of procedural justice) is not.

6. Discussion

The surge in AI adoption across industries brings both opportunities and challenges for firms, which must find ways to use AI to strengthen competitive advantages while maintaining fair and trustworthy relationships with their supply chain partners. Our study addresses this tension by demonstrating that responsible AI may serve not only as a technical tool but also as a mechanism conducive to perceived justice in supply chain relationships. By embedding principles of accountability, responsibility, and transparency into AI use, firms can actively shape their partners' perceptions of fairness, which in turn acts as a catalyst for competitive advantages. Importantly, this effect is unlikely to arise from fairness perceptions alone; rather, it is realized through concrete cooperative behaviors between supply chain partners that affect costs, responsiveness, and relational investments.

First, our results indicate that responsible AI is positively related to distributive justice by optimizing the equity of resource allocation. Grounded in OJT, partners assess fairness by comparing their input-to-outcome ratio to that of others (Adams, 1965). Responsible AI facilitates this process by using objective data to allocate orders, prices, and resources (Floridi et al., 2018). Further, AI's capacity for dynamic sensing enables needs-based allocation during shortages, ensuring partners receive support based on urgency. When partners perceive these outcomes as equitable and derived from objective criteria, they are less inclined to engage in defensive bargaining or opportunistic hedging. In turn, these fairness perceptions can improve competitive advantages through concrete cooperative behaviors (Chen et al., 2022). Perceived distributive fairness reduces costly bargaining and defensive monitoring, thereby lowering transaction costs. It can also motivate partners to reciprocate through stronger cooperation in operations, such as giving the focal firm priority access during shortages or being more willing to accommodate temporary adjustments in allocation. Through lower relational frictions and more reliable partner support, distributive justice helps translate responsible AI into stronger cost efficiency and market responsiveness, which are reflected in superior competitive advantages (Branco and Rodrigues, 2006).

Second, responsible AI acts as a powerful driver of procedural justice. Our findings suggest that ethical AI practices operationalize Leventhal's (1980) core justice criteria, specifically consistency, unbiasedness, and correctability, into supply chain governance. By replacing opaque manual decisions with transparent algorithms (explainable AI) and providing mechanisms for feedback (voice), responsible AI signals that the decision-making structure is legitimate and respectful (Arrieta et al., 2020). Unlike favorable outcomes, which may fluctuate, a fair process reduces uncertainty and creates willingness to rely on the partner. This willingness is strategically consequential because it enables cooperation beyond contractual minimums. When partners believe that decision rules are consistent and unbiased, they are less likely to engage in opportunistic behaviors or conflict, reducing the need for costly safeguards and lowering transaction costs. At the same time, fair procedures encourage knowledge sharing, relationship-specific investment, and voluntary cooperation, which improve product quality and help firms respond more agilely to market changes (Huo et al., 2016; Liu et al., 2012). These behaviors improve forecast accuracy, reduce the costs of coordination and conflict resolution, and support innovation and adaptability. Over time, such relational assets become difficult for competitors to imitate, thereby converting procedural justice into sustainable competitive advantages and, ultimately, stronger relative

financial performance.

Finally, our study uncovers a critical asymmetry in how SCC affects these two justice pathways. As hypothesized, high SCC weakens the mediating role of distributive justice. In multitier, information-poor networks, the social comparison process needed to assess distributive equity is disrupted by attributional ambiguity (Wang et al., 2022). Partners simply cannot see the full picture to verify whether their slice of the pie is fair. However, contrary to the expectation that SCC would universally degrade perceptions of fairness, we found that perceptions of procedural justice remain robust. This finding aligns with the uncertainty management perspective in the justice literature (Lind, 2001), which suggests that under conditions of high uncertainty (e.g., SCC), where outcomes are hard to measure, actors rely heavily on procedural information as a cognitive anchor. In this case, SCC increases the need for heuristics. Because the features of responsible AI (e.g., transparency and explanation protocols) are directly experienced by the partner regardless of network noise (Wagner and Neshat, 2010), procedural justice retains its effectiveness. This suggests that in complex environments, the fairness of the process becomes the dominant signal of trustworthiness, overriding the ambiguity of outcomes.

6.1. Theoretical implications

Our study offers three distinct contributions to the literature on AI-enabled supply chain management and interorganizational justice. First, we extend the antecedents of justice by conceptualizing responsible AI as a novel governance design. Prior literature has largely focused on traditional governance mechanisms such as formal contract complexity (Poppo and Zhou, 2014), relational norms (Brito and Miguel, 2017), and power dynamics (J. Hoppner et al., 2014) as drivers of justice perceptions. We identify responsible AI as a form of algorithmic governance that operationalizes justice criteria. Specifically, we demonstrate how technical attributes such as bias mitigation and explainability function as the digital equivalents of Leventhal's (1980) rules of unbiasedness and accuracy. This bridges the gap between AI ethics and supply chain governance, showing that ethical algorithms not only ensure compliance but actively construct the psychosocial infrastructure (e.g., partners' psychological safety and reduced fear of exploitation) necessary for deep collaboration.

Second, we answer the call to link interorganizational justice to strategic performance outcomes. As noted by Narasimhan et al. (2013) and Alghababsheh et al. (2023), existing research has primarily focused on attitudinal outcomes (e.g., satisfaction), with limited evidence of tangible performance. Our study empirically supports the justice-competitive advantage link and clarifies why justice has strategic value in AI-enabled supply chains. Specifically, the value of distributive and procedural justice lies not only in more favorable partner evaluations but also in the cooperative behaviors that follow from those evaluations. Distributive justice can reduce costly bargaining and defensive monitoring and encourage stronger partner support in times of scarcity, whereas procedural justice can reduce opportunistic behaviors and conflict while promoting knowledge sharing, relationship-specific investment, and voluntary cooperation. In this way, justice helps firms lower transaction costs, improve responsiveness, and strengthen longer-term collaborative advantages. This distinguishes responsible AI from a purely efficiency-oriented view of AI adoption by showing that part of its value is relational and is realized through fair supply chain exchanges.

Third, we refine the boundary conditions of justice by revealing the enduring value of procedural justice under conditions of complexity. Previous studies on environmental uncertainty have yielded inconsistent results regarding the efficacy of justice (Cheng et al., 2020; Kumar et al., 1995). Our findings resolve some of these tensions by highlighting the asymmetric sensitivity of justice dimensions. We show that while SCC acts as a barrier to outcomes-based (distributive) fairness because of information asymmetry, it does not diminish the value of process-based

(procedural) fairness. This pattern is consistent with fairness heuristic theory (Lind, 2001), which suggests that when outcomes are difficult to observe or evaluate, actors rely on procedural cues as a practical shortcut for judging whether another party can be trusted. In AI-enabled supply chains, this means that transparent, consistent, and explainable procedures may carry greater weight in complex networks. Importantly, this logic also points to a potential boundary of procedural justice in algorithmic governance: under high complexity, partners may place substantial weight on visible procedural features of AI even when they have limited ability to verify whether allocation outcomes are equally favorable. Thus, our findings suggest that procedural justice is especially robust under complexity, but they also imply that future research should examine when AI-based process transparency is aligned with distributive fairness and when the two may diverge.

6.2. Practical implications

Our findings offer actionable guidance for managers navigating the intersection between AI governance and supply chain relationship management. Critically, our results suggest that the competitive value of responsible AI stems not only from its technical deployment but also from its ability to shape partners' perceptions of justice. Therefore, managers should shift their focus from AI implementation to managing justice perceptions.

First, managers should ensure that fairness is visible to ensure it is perceived by partners. Our study confirms that responsible AI only fosters competitive advantages when it successfully translates into perceived distributive and procedural justice. A technically unbiased algorithm that operates as a black box fails to contribute to this value. Therefore, managers should implement visible justice mechanisms. Instead of just optimizing backend allocation, firms should provide partners with explanatory algorithmic reports. For example, if an AI system reduces a supplier's order volume, the system should automatically generate a plain language explanation with objective data. Moreover, managers could establish AI complaint channels through which partners can flag potential errors in AI judgments. This operationalizes the correctability criterion of procedural justice, turning a passive technological transaction into an interactive, respectful governance process.

Our results also offer guidance for adopting scenario-specific strategies based on SCC. Recognizing that SCC acts as a boundary condition, we propose a differentiated approach for managing AI governance in simple versus complex networks. In low-complexity environments (e.g., direct Tier 1 relationships), managers should pursue both equitable outcomes (distributive) and transparent processes (procedural) because these are easily verified by partners. In this case, AI systems should provide detailed data showing how a partner's treatment compares with the market average, reinforcing the perception of equity. In high-complexity environments (e.g., multitier, global networks), instead of endeavoring to prove that every allocation is fair, managers should pivot strategically to ensuring the consistency, accessibility, and transparency of processes. For example, implementing a unified, blockchain-backed AI interface that guarantees that the same evaluation rules are applied to a number of suppliers is more effective than trying to justify individual order splits. By anchoring partners' trust in the process (which remains visible) rather than the outcome (which is obscure), firms can maintain collaboration, even in chaotic environments. At the same time, managers should not treat procedural transparency as a substitute for substantive fairness. In complex networks, explainability should be paired with auditability, appeal mechanisms, and periodic reviews of allocation outcomes. This strategic pivot is particularly relevant for large-scale manufacturing firms that manage extensive supplier bases, where verifying individual outcomes is costly. In contrast, for smaller IT service providers, where relationships are more direct, emphasizing outcome equity remains a viable and effective strategy.

7. Conclusion

Drawing on OJT, we empirically examined the relational mechanisms linking responsible AI to competitive advantages. Our analysis of survey data from 218 Chinese firms reveals that responsible AI is positively associated with both distributive and procedural justice, which subsequently serve as key mediators in fostering competitive advantages. Crucially, our findings highlight a divergence in how SCC conditions these effects: while high SCC weakens the mediating role of distributive justice, the mediating role of procedural justice remains robust. Practically, this suggests that in complex supply networks, managers should prioritize the transparency and consistency of decision-making processes because fair procedures offer a more stable mechanism for maintaining cooperation than outcome-based equity alone.

Our study has several limitations that suggest directions for future research. First, our cross-sectional design could not capture the temporal evolution of justice perceptions; future researchers could use longitudinal designs to validate causal directions. Second, although our theorizing suggests that supply chain justice improves competitive advantages through cooperative behaviors between partners, we did not directly measure these intermediate behaviors in our survey. Future research could incorporate more fine-grained measures of interorganizational behaviors to uncover how justice perceptions are translated into competitive advantages. Third, while we adapted established scales, our items for supply chain justice were relatively general. Future researchers could develop context-specific scales that explicitly reference AI-driven tasks (e.g., algorithmic inventory allocation) to capture finer-grained fairness perceptions. Similarly, our theoretical model prioritized distributive and procedural justice as the primary structural mechanisms. However, given the multidimensional nature of supply chain justice, future researchers could extend this framework to include interactional and informational justice to capture the interpersonal and communicative differences of AI-enabled collaboration. Additionally, future works could broaden the responsible AI construct by explicitly incorporating accountability and complement our complexity measures with objective structural metrics. Finally, the use of qualitative methods such as case studies would provide deeper insights into the organizational barriers and contextual differences of responsible AI adoption that quantitative surveys may overlook.

CRedit authorship contribution statement

Lixu Li: Writing – original draft, Conceptualization. **Ruitian Shi:** Data curation. **Ruijie Zhang:** Formal analysis. **Lujie Chen:** Writing – review & editing, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

No.

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Appendix

Responsible AI (CR = 0.812; AVE = 0.521; Cronbach's α = 0.809)	<i>When using AI...</i> ...we continuously attempt to reduce algorithm biases (0.722). ...we are impartial and avoid unfair discrimination (0.759). ...we maintain privacy protection (0.633). ...AI-based processes are transparent to all stakeholders (0.766).
Distributive justice (CR = 0.799; AVE = 0.570; Cronbach's α = 0.796)	- Our gain consistent with effort and investment (0.774). - Our gain commensurate with role and responsibilities (0.699). - Our gain proportionate to our performance (0.789).
Procedural justice (CR = 0.822; AVE = 0.606; Cronbach's α = 0.818)	- We are fair in our dealings with supply chain members (0.769). - We fully explained the decision-making criteria to supply chain members (0.749). - We applied consistent decision-making criteria when dealing with supply chain members (0.816).
Competitive advantages (CR = 0.820; AVE = 0.602; Cronbach's α = 0.817)	- Our company's return on investment (ROI) is better compared to other companies in the same industry (0.773). - Our company's return on equity (ROE) is better compared to other companies in the same industry (0.774). - Our company's return on asset (ROA) is better compared to other companies in the same industry (0.781).
Supply chain complexity (CR = 0.879; AVE = 0.709; Cronbach's α = 0.876)	- Our supply chain is very complex (0.909). - Our supply chain involves a lot of players (e.g., suppliers, logistics service providers) and/or a lot of logistics/transportation transactions (0.745). - Our supply chain is a quite intricate network (0.864).

Note: Numbers in parentheses are factor loadings

Fig. A1. Assessment of measurement items.

Data availability

Data will be made available on request.

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